WS01 - Heat, temperature, thermal energy

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Distinguish temperature, internal energy and heat; predict the direction of spontaneous heat transfer; explain why sample size matters.

WS01-Q01 (2 points) Which statement best defines heat? A. The temperature of an object. B. Energy transferred because of a temperature difference. C. The number of particles. D. Energy permanently stored as a substance.

WS01-Q02 (2 points) A 42 °C block touches a 17 °C block. State the initial direction of net heat transfer and its cause.

WS01 - Heat, temperature, thermal energy

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS01-Q03 (2 points) A 40 g water sample and a 400 g water sample are both at 22 °C. Compare their temperatures and their total internal energies under the same phase and conditions.

WS01-Q04 (2 points) Rewrite this sentence: “The cup contains heat, measured in degrees Celsius.”

WS01 - Heat, temperature, thermal energy

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS01-Q05 (3 points) No work occurs. A sample receives 460 J and transfers 125 J out. Find its net internal-energy change. Show the subtraction.

WS01-Q06 (3 points) A student says a 90 °C drop must have more total internal energy than a large 40 °C bucket of the same water. Explain why the stated temperatures do not prove the claim.

WS01 - Heat, temperature, thermal energy

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS01-Q07 (3 points) Draw two objects, each at 25 °C, in thermal contact. Label the net heat transfer and state whether their particles have stopped moving.

WS01-Q08 (3 points) Challenge: A metal piece loses 240 J while water gains 205 J. No work occurs. How much energy goes to other surroundings? Explain why this need not violate conservation.

WS02 - Thermodynamic systems and surroundings

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Choose a useful system boundary; classify open, closed and ideal isolated systems; distinguish matter exchange from energy exchange.

WS02-Q01 (2 points) Define the system and surroundings if the question asks only about the water in an open cup.

WS02-Q02 (2 points) A sealed, leak-free bottle warms in sunlight. Is it open, closed or isolated? Explain.

WS02 - Thermodynamic systems and surroundings

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS02-Q03 (2 points) An uncovered dish loses water by evaporation. Classify the water system and identify what crosses its boundary.

WS02-Q04 (2 points) A model container exchanges neither matter nor energy. Give its system type and one reason a real container may only approximate this.

WS02 - Thermodynamic systems and surroundings

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS02-Q05 (3 points) A tank initially holds 220 g of water. Add 35 g and remove 18 g. Find the final mass and classify this during the exchange.

WS02-Q06 (3 points) Draw a boundary around a sealed bottle plus its contents. Mark an energy arrow entering and state whether any matter arrow is allowed under the leak-free assumption.

WS02 - Thermodynamic systems and surroundings

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS02-Q07 (3 points) A closed gas cylinder has a moving piston. Can the gas exchange energy by work without exchanging matter? Explain using the boundary.

WS02-Q08 (3 points) Challenge: A student calls an open, insulated cup isolated. Identify two separate checks needed before accepting that claim.

WS03 - Intensive and extensive properties

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Classify intensive and extensive properties; apply the split/combine test; use density and mass consistently.

WS03-Q01 (2 points) Classify mass and temperature as extensive or intensive.

WS03-Q02 (2 points) Classify volume and density for uniform samples at the same state.

WS03 - Intensive and extensive properties

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS03-Q03 (2 points) Two identical 30 °C portions are combined without energy loss or phase change. Choose the final temperature: A. 15 °C B. 30 °C C. 60 °C. Explain.

WS03-Q04 (2 points) A homogeneous 90 g sample is divided into three equal portions with no state change. What is each mass? What happens to its density?

WS03 - Intensive and extensive properties

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS03-Q05 (3 points) A liquid sample has mass 63 g and volume 70 mL. Calculate its density.

WS03-Q06 (3 points) Double the sample in Q05 using an identical portion. Give the new mass, volume and density.

WS03 - Intensive and extensive properties

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS03-Q07 (3 points) A student argues pressure is extensive because it can change when a gas is compressed. Explain the mistake.

WS03-Q08 (3 points) Challenge: Under a stated consistent energy reference, one uniform portion has U=600 J, mass 20 g and temperature 28 °C. Combine three identical portions without other effects. Give total U, total mass and temperature.

WS04 - Thermal equilibrium and zeroth law

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Use thermal equilibrium and the zeroth law; distinguish equilibrium from equal energy; explain how a thermometer obtains a useful reading.

WS04-Q01 (2 points) In one sentence, define thermal equilibrium between two objects in thermal contact.

WS04-Q02 (2 points) A is in equilibrium with reference C. B is also in equilibrium with unchanged C. What follows about A and B, and which law says so?

WS04 - Thermal equilibrium and zeroth law

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS04-Q03 (2 points) Can a 20 g water sample and a 200 g water sample at the same temperature be in thermal equilibrium? Explain.

WS04-Q04 (2 points) A probe starts colder than a sample. Why might an immediate reading be too low?

WS04 - Thermal equilibrium and zeroth law

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS04-Q05 (3 points) Two equal water masses start at 16.0 °C and 28.0 °C. Assume equal constant specific heats, no work, no phase change, no external exchange and negligible container energy. Find the common final temperature.

WS04-Q06 (3 points) Read this simulated dataset: time (min) 0, 2, 4, 6; object A (°C) 36, 32, 29, 27; object B (°C) 18, 22, 25, 27. At which listed time is thermal equilibrium indicated? State the earlier net-flow direction.

WS04 - Thermal equilibrium and zeroth law

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS04-Q07 (3 points) Explain two reasons “the particles have stopped” is a poor description of equilibrium.

WS04-Q08 (3 points) Challenge: A 10 g sample at 80 °C and a 100 g sample at 20 °C are combined. A student uses 50 °C by averaging. Without finding the true final temperature, explain why that shortcut is unjustified and say which initial temperature the final value should be closer to if both are water and the system is insulated.

WS05 - Celsius, Fahrenheit, Kelvin

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Convert °C, °F and K; distinguish absolute readings from changes; select the appropriate scale for a calculation.

WS05-Q01 (2 points) Convert 15.0 °C to K. Round to one decimal place.

WS05-Q02 (2 points) Convert 310.0 K to °C. Round to one decimal place.

WS05 - Celsius, Fahrenheit, Kelvin

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS05-Q03 (2 points) Convert 10.0 °C to °F.

WS05-Q04 (2 points) The temperature increases by 12.0 °C. What is the increase in K? Explain why no offset is needed.

WS05 - Celsius, Fahrenheit, Kelvin

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS05-Q05 (3 points) Convert 68.0 °F to °C. Show the operation order.

WS05-Q06 (3 points) A sample cools from 31.0 °C to 23.0 °C. Give final minus initial in °C, K and °F.

WS05 - Celsius, Fahrenheit, Kelvin

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS05-Q07 (3 points) A learner enters 25 into an ideal-gas calculation that requires absolute temperature because the gas is at 25 °C. What reading should be used, and why? Give the unrounded conversion.

WS05-Q08 (3 points) Challenge: At what temperature are the numerical Celsius and Fahrenheit readings equal? Use $x=1.8x+32$ and solve for x.

WS06 - Heat-energy units and numerical conversions

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Convert heat-related energy units; distinguish power from energy; use unit cancellation to check a calculation.

WS06-Q01 (2 points) Convert 2.70 kJ to J.

WS06-Q02 (2 points) Convert 840 J to kJ.

WS06 - Heat-energy units and numerical conversions

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS06-Q03 (2 points) Using 1 cal=4.184 J, convert 15.0 cal to J. Give three significant figures.

WS06-Q04 (2 points) Complete the statements: 1 food Calorie equals ___ small calories. A watt measures ___, not total energy.

WS06 - Heat-energy units and numerical conversions

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS06-Q05 (3 points) A constant transfer rate is 4.0 W for 2.0 min. Find the transferred energy. Give two significant figures.

WS06-Q06 (3 points) Convert 3.00 food Calories to kJ. Use the thermochemical calorie and give three significant figures.

WS06 - Heat-energy units and numerical conversions

Name: _____ Date: _____

20–25 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

WS06-Q07 (3 points) A student claims that 1000 J always produces a 1000 °C rise. Explain what is wrong and name two sample properties needed for a simple single-phase calculation.

WS06-Q08 (3 points) Challenge: An illustrative lamp supplies 90 W for 10 min. Suppose only 5.0% of that energy reaches the water. Find the supplied energy and the energy reaching the water. This is a math model, not a competition-lamp specification.

D01 - Prerequisite diagnostic

Name: _____ Date: _____

15 minutes total | 20 points | Page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Check signed arithmetic, decimals, unit conversions, one-variable algebra, geometry and table reading.

D01-Q01 (2 points) Calculate $48-65$.

D01-Q02 (2 points) Calculate 0.25×120.

D01 - Prerequisite diagnostic

Name: _____ Date: _____

15 minutes total | 20 points | Page 2 of 4

Difficulty: Beginner through application; final question is a challenge

D01-Q03 (2 points) Convert 3 minutes to seconds; use 60 seconds per minute.

D01-Q04 (2 points) Solve $5x=35$.

D01 - Prerequisite diagnostic

Name: _____ Date: _____

15 minutes total | 20 points | Page 3 of 4

Difficulty: Beginner through application; final question is a challenge

D01-Q05 (3 points) Solve $y=2x+6$ for x when $y=20$.

D01-Q06 (3 points) Find the area of a rectangle 8 cm long and 3 cm wide.

D01 - Prerequisite diagnostic

Name: _____ Date: _____

15 minutes total | 20 points | Page 4 of 4

Difficulty: Beginner through application; final question is a challenge

D01-Q07 (3 points) A table lists time (min) 0, 2, 4 and a reading (°C) 20, 23, 25. What is the reading at 2 min, and how much does it change from 0 to 4 min?

D01-Q08 (3 points) Find 10% of 250, then explain whether the result should be greater or less than 250.